

Music Playlist Manager in C++

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#include <iostream>
#include <string>
using namespace std;

class Song {
public:
    string title;
    Song* next;
    Song* prev;

    Song(string name) {
        title = name;
        next = NULL;
        prev = NULL;
    }
};

class Playlist {
private:
    Song* head;
    Song* tail;
    Song* current;

public:
    Playlist() {
        head = NULL;
        tail = NULL;
        current = NULL;
    }

    void addSong(string title) {
        Song* newSong = new Song(title);

        if(head == NULL) {
            head = tail = current = newSong;
        }
        else {
            tail->next = newSong;
            newSong->prev = tail;
            tail = newSong;
        }

        cout << "Song Added Successfully!\n";
    }

    void displayPlaylist() {
        if(head == NULL) {
            cout << "Playlist is Empty!\n";
            return;
        }

        Song* temp = head;

        cout << "\n===== PLAYLIST =====\n";

        while(temp != NULL) {
            cout << "- " << temp->title << endl;
            temp = temp->next;
        }
    }

    void playCurrentSong() {
        if(current == NULL) {
            cout << "No Song Available!\n";
            return;
        }

        cout << "Now Playing: " << current->title << endl;
    }

    void nextSong() {
        if(current == NULL || current->next == NULL) {
            cout << "No Next Song!\n";
            return;
        }

        current = current->next;
        playCurrentSong();
    }
};
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}

void previousSong() {
    if(current == NULL || current->prev == NULL) {
        cout << "No Previous Song!\n";
        return;
    }

    current = current->prev;
    playCurrentSong();
}

void removeSong(string title) {
    if(head == NULL) {
        cout << "Playlist is Empty!\n";
        return;
    }

    Song* temp = head;

    while(temp != NULL && temp->title != title) {
        temp = temp->next;
    }

    if(temp == NULL) {
        cout << "Song Not Found!\n";
        return;
    }

    if(temp == head)
        head = head->next;

    if(temp == tail)
        tail = tail->prev;

    if(temp->prev != NULL)
        temp->prev->next = temp->next;

    if(temp->next != NULL)
        temp->next->prev = temp->prev;

    if(current == temp)
        current = head;

    delete temp;

    cout << "Song Removed Successfully!\n";
}

void menu() {
    int choice;
    string songName;

    do {
        cout << "\n===== MUSIC PLAYLIST MENU =====\n";
        cout << "1. Add Song\n";
        cout << "2. Remove Song\n";
        cout << "3. Display Playlist\n";
        cout << "4. Play Current Song\n";
        cout << "5. Next Song\n";
        cout << "6. Previous Song\n";
        cout << "7. Exit\n";
        cout << "Enter Your Choice: ";
        cin >> choice;

        cin.ignore();

        switch(choice) {
            case 1:
                cout << "Enter Song Name: ";
                getline(cin, songName);
                addSong(songName);
                break;

            case 2:
                cout << "Enter Song Name to Remove: ";
                getline(cin, songName);
                removeSong(songName);
                break;

            case 3:
                displayPlaylist();

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        break;

    case 4:
        playCurrentSong();
        break;

    case 5:
        nextSong();
        break;

    case 6:
        previousSong();
        break;

    case 7:
        cout << "Exiting Playlist...\n";
        break;

    default:
        cout << "Invalid Choice!\n";
    }
} while(choice != 7);
};

int main() {
    Playlist playlist;

    playlist.menu();

    return 0;
}

```